

AN EXECUTIVE DATA ANALYSIS REPORT

State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

Key insights from JetBrains' survey of 23,262 developers worldwide. For technology strategy, hiring priorities, and developer experience.

23,262

developers surveyed

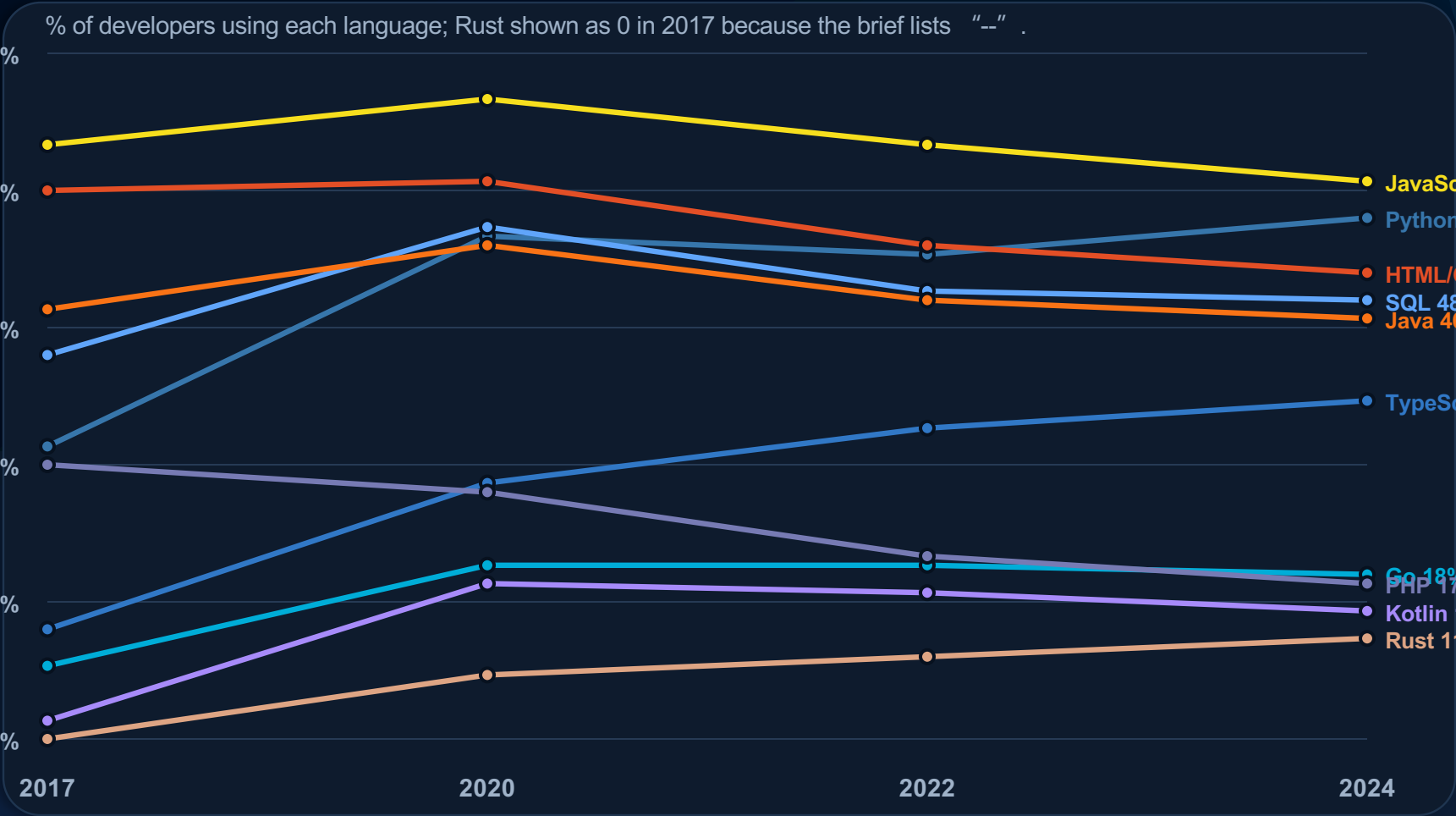
2017–2024

8-year language trends

3 lenses

usage, love, domain fit

Python Closes the Gap: 8 Years of Language Usage Shifts



EXECUTIVE READ

● Python +25 pts

● TypeScript x3

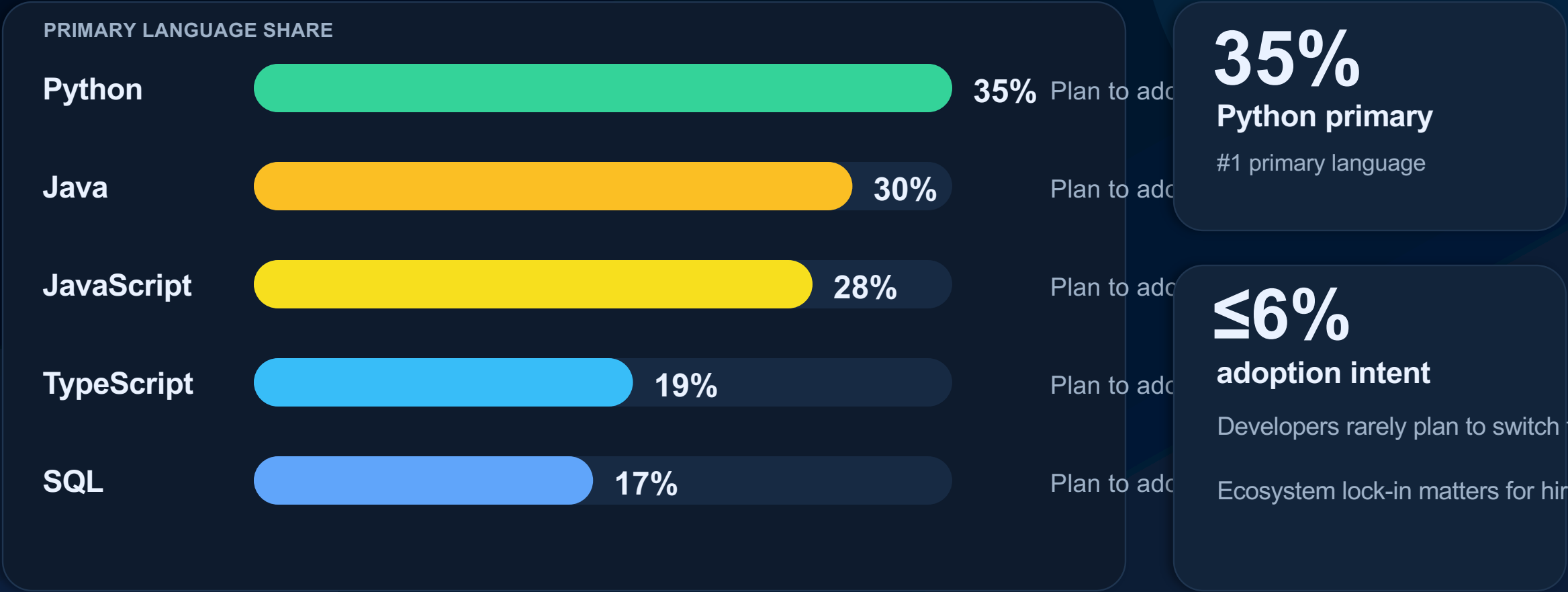
● JavaScript peak → decline

● PHP -13 pts

The top-5 landscape is reshuffling as AI/c

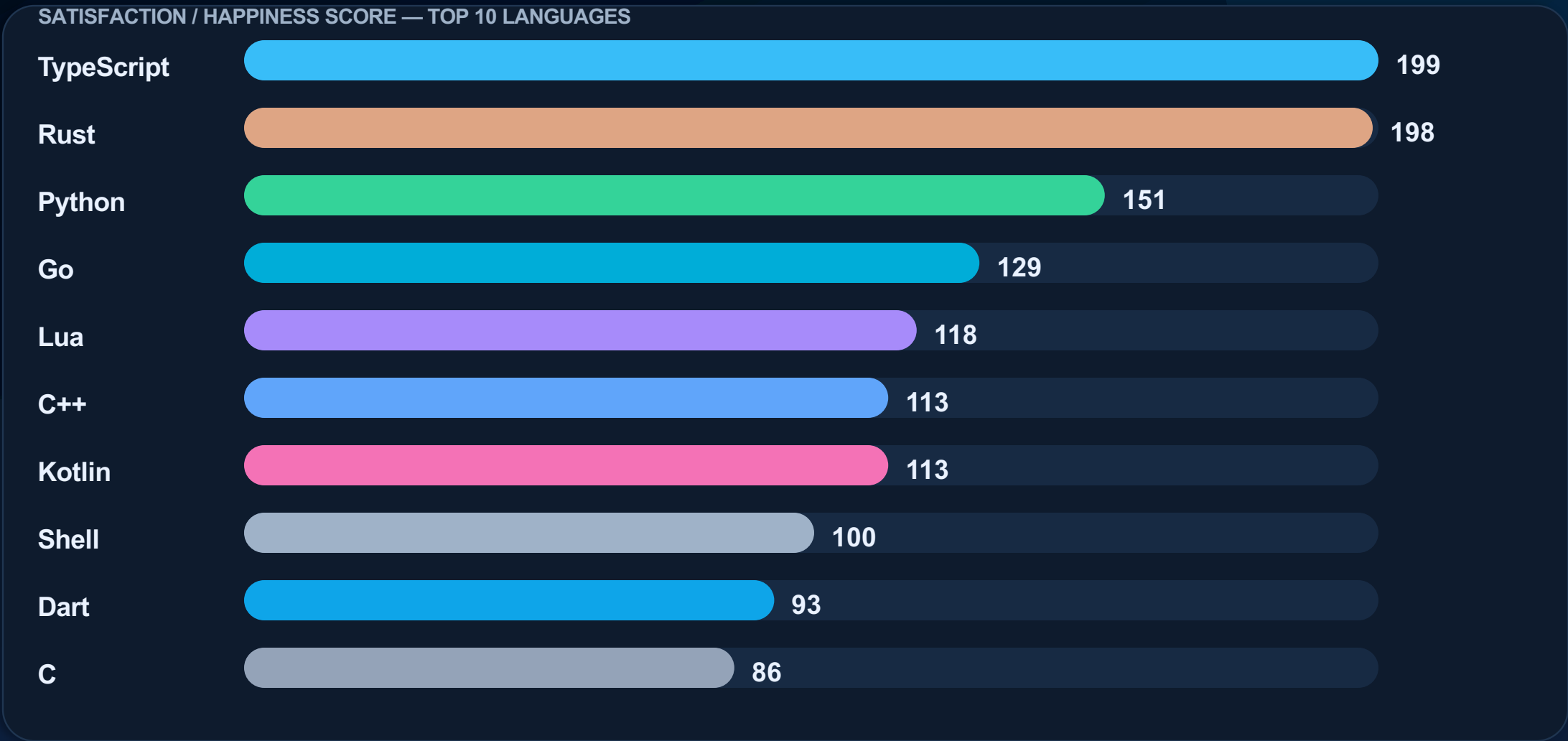
JavaScript remains most used at 61%, but Python has nearly closed the gap at 57%.

Python Is Now the #1 Primary Language at 35%



Comfort-zone effect: primary stack choices are sticky even when broader usage patterns shift.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



TypeScript and Rust are effectively tied, with Python a clear third.

Signal: type safety + modern tooling

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

DOMAIN	LEADING LANGUAGE	SHARE
AI / Machine Learning	Python	31%
Data Science / Analytics	Python	28%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
User Interface	JavaScript	58%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	33%

Strategy lens
Choose by domain fit, not one universal winner

Python dominates AI/data; TypeScript and JavaScript own UI; Java leads services; C++ leads systems work.

What This Means for Your Technology Strategy in 2025

The action plan: align language bets to business domains, talent markets, and developer experience.

1

Invest in Python for AI/ML

Python leads AI/ML at 31% and data science at 28%.

2

Standardize TypeScript for UI/full-stack

TypeScript leads UI at 62% and tops satisfaction at 199.

3

Use Rust + Go as infrastructure growth plays

Rust continues climbing to 11%; Go stabilized near 18%.

4

Rebalance legacy hiring risk

PHP fell from 30% to 17%; Ruby fell from 10% to 4%.

5

Treat developer experience as retention strategy

High satisfaction clusters around type safety and modern tooling.

Bottom line: optimize for domain advantage and retention — not language popularity alone.